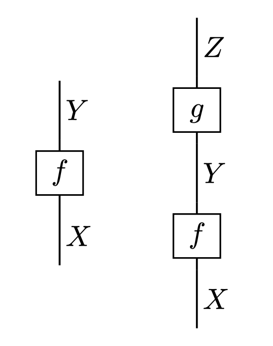
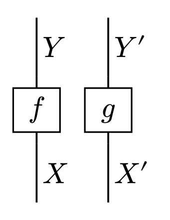
drawstring gallery

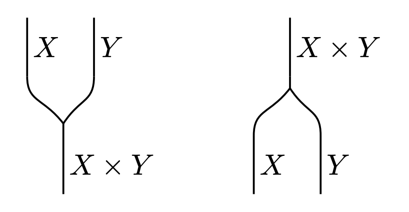
A kernel $f : X \rightsquigarrow Y$, and sequential composition:



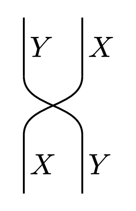
Parallel composition:



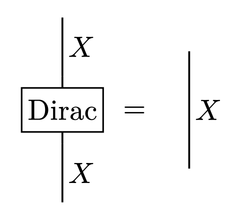
A product wire $X \times Y$ may be drawn as two wires, and back:



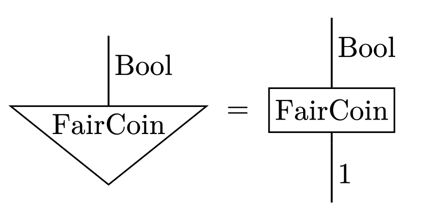
The swap:



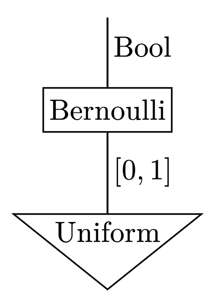
The Dirac kernel is the bare wire:



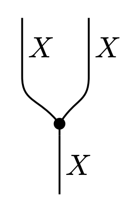
A state is a kernel with trivial input:



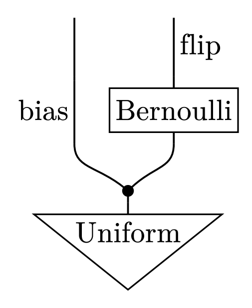
A little probabilistic program:



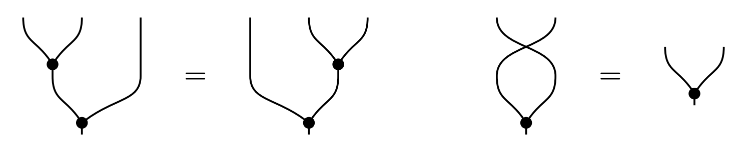
Copying:



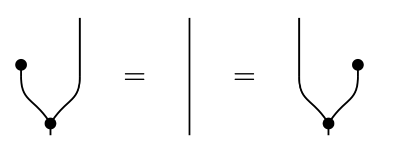
Sampling a bias, then flipping a coin with it:



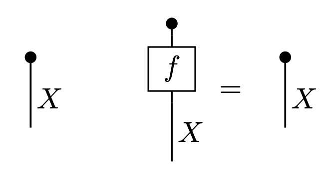
Coassociativity and cocommutativity of copying:



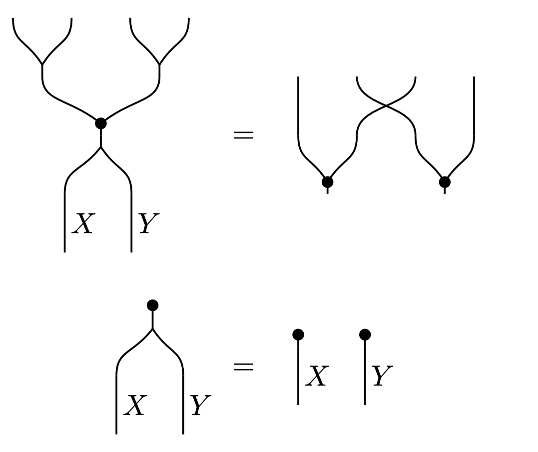
Counitality:



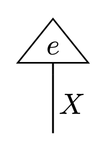
Discarding, and the fact that every kernel is discarded to nothing:



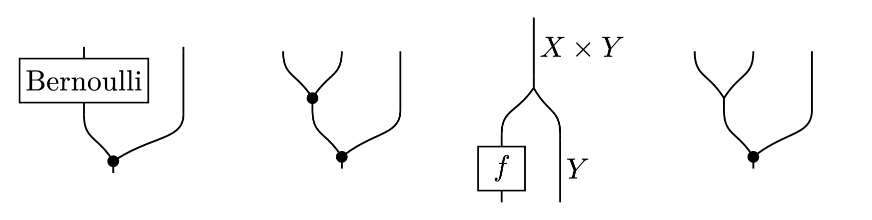
Copying and discarding a product:



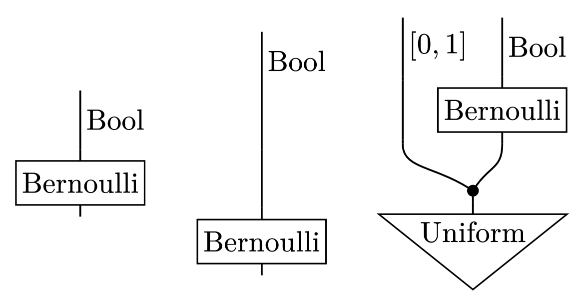
An effect (the mirror image of a state):



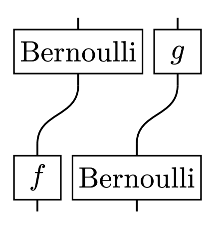
The arms of `copy`, `unbundle` and `bundle` are flexible: each runs straight from its dot to wherever the neighbouring layer needs it.



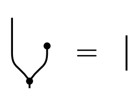
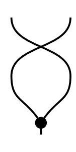
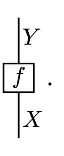
Plain wires are flexible too. They follow the box they stand on, so none of these needs a connector band:



A wire whose two ends are held at different positions bends by itself, with no band inserted:

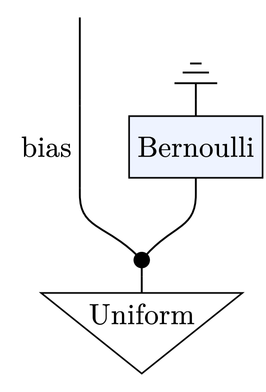


Inline diagrams

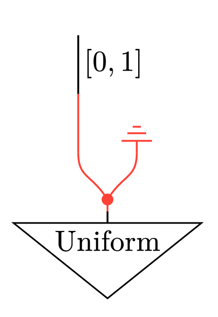
The copy map and the discard map satisfy  , so every object is a comonoid. At the default unit the same term,  , is too tall for a line of text, so inline diagrams should pass a smaller `unit`, and a smaller label size when they carry labels:  .

Style overrides

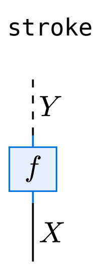
A style can be set for the whole diagram, through the `style:` argument of `string-diagram`. Element groups inherit the root `stroke` and `fill`, and partial strokes fold as in cetz:



It can be set for a sub-diagram, through `styled` or the `style:` argument of `serial` and `parallel`:

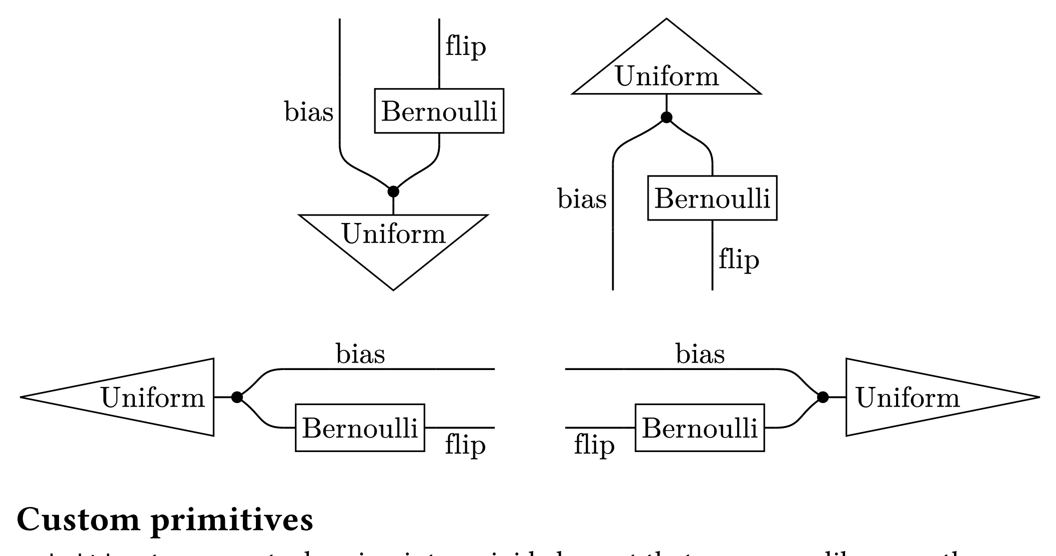


Or for a single element, through its own `stroke` and `fill` arguments:



Reading direction

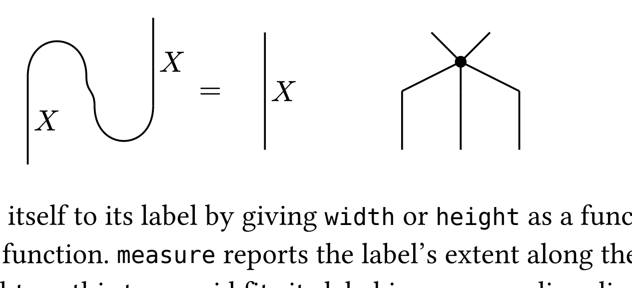
The style key `direction` reads the same diagram bottom to top (the default), top to bottom, left to right, or right to left. Read sideways, a wire label with `side: "right"` sits below its wire and one with `side: "left"` above it.



Custom primitives

`primitive` turns a cetz drawing into a rigid element that composes like any other. The drawing function receives the resolved style and the element's geometry, and draws in units with the origin at the element's bottom-left corner.

The snake equation of a compact closed category, and a spider:



An element can size itself to its label by giving `width` or `height` as a function of the style and a `measure` function. `measure` reports the label's extent along the element's own width and height, so this trapezoid fits its label in every reading direction: